An introduction to

Suti

Standardized Utilization of Transport Information

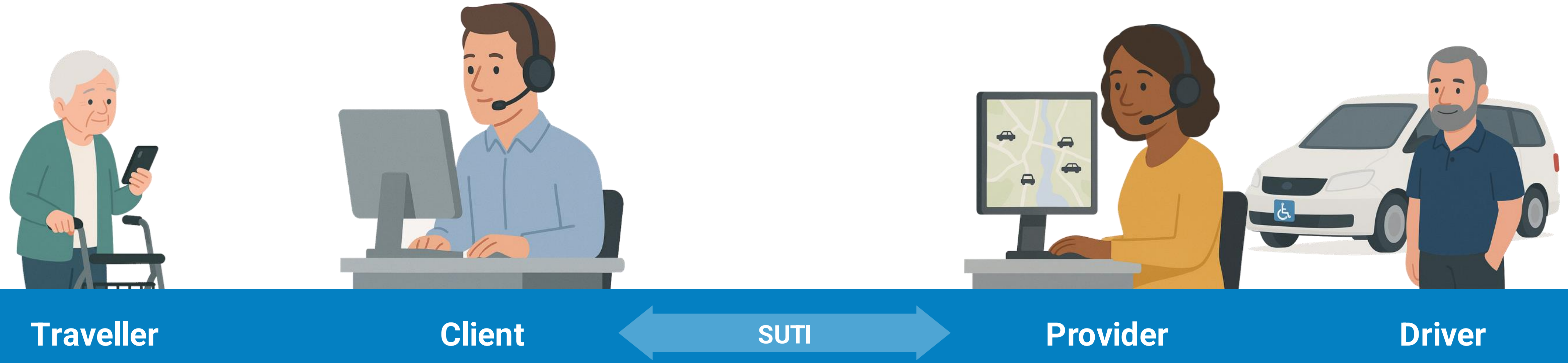
What is SUTI used for?

In Demand Responsive Transport (DRT) the following actors are involved:

Travellers in need of transport – the Demand.
Clients that aggregate transport need and create bookings to satisfy those needs, thereby acting on behalf of Travellers. A typical Client is a Public Transport Authority.
Providers with **Drivers** and Vehicles that carry out the transport needs – the Response. A typical Provider is a Taxi Company.

SUTI (Standardized Utliization of Transport Information) is a **technical communication protocol**. It is used for communicating between a Client’s planning and booking system and a Provider’s vehicle dispatch system.

The standard is used by hundreds of DRT services, handling more than 30 million orders yearly in the Nordic region.



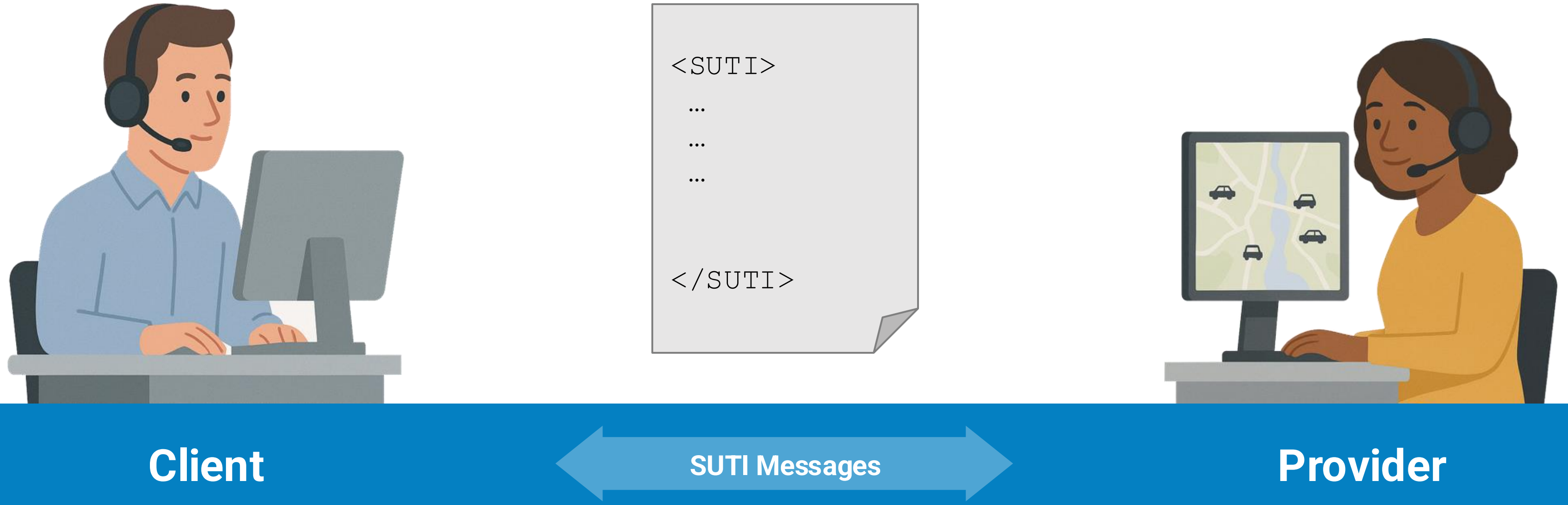
SUTI communicates using Messages

SUTI defines a number of message types that can be sent between a Client and a Provider, over a **SUTI Link**. A message will contain either a request, or a response to a request. The standard covers relevant Use Cases before, during and after transport.

SUTI messages are formatted as XML or JSON. The standard doesn't prescribe a specific communication method. The following methods are common among the numerous different implementations:

- **Port-to-port** communication (physcial or logical over the internet)
- **HTTP-communication with a single endpoint**, set up by the Client, that the Provider calls to send messages and polls to receive messages.
- **HTTP-communication with dual endpoints**, set up by the Client and Provider respectively.

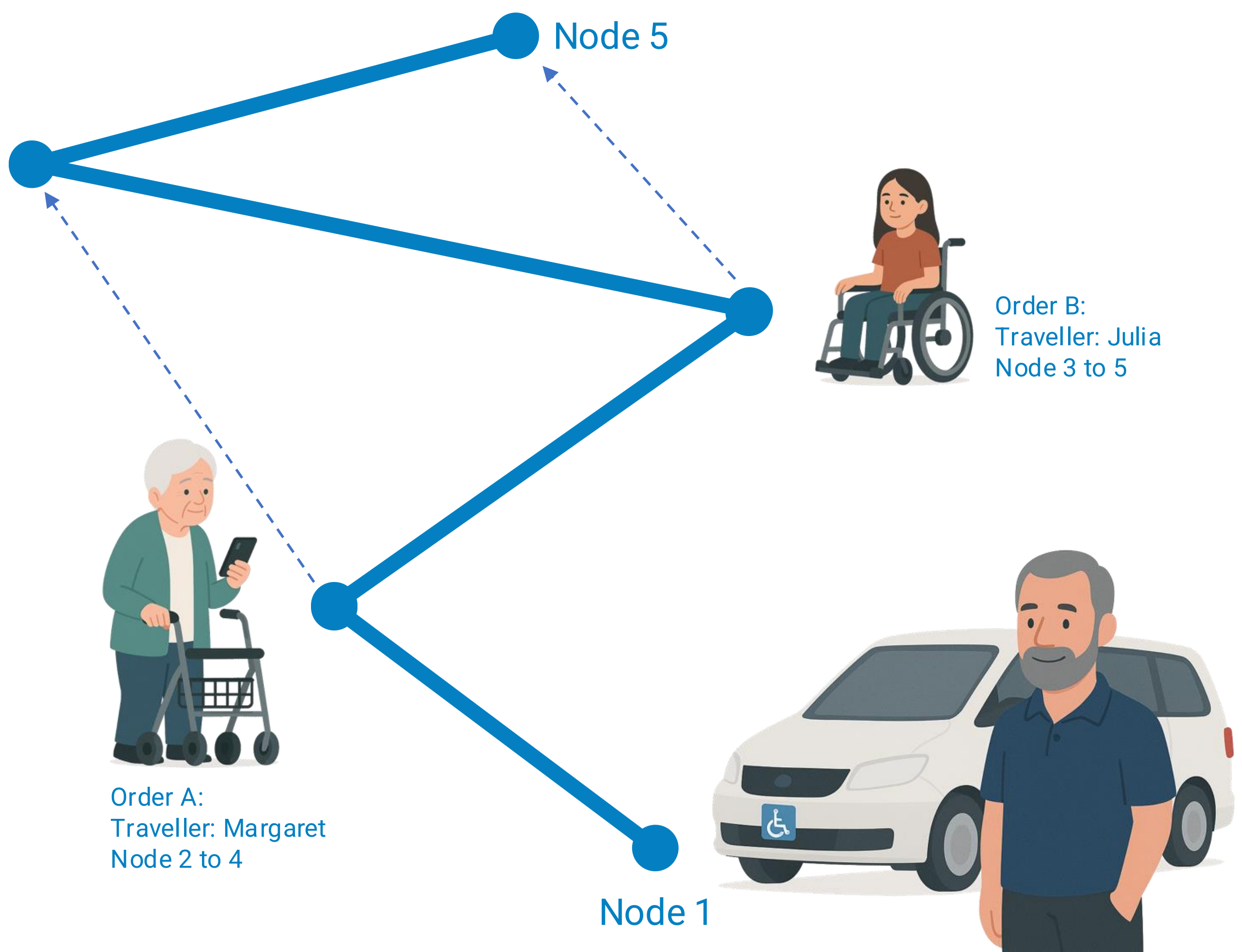
Each Client defines a preferred communication method and most Provider softwares support all of them.



Order Flows

A SUTI implementation defines how Orders are built, communicated and maintained.

In most cases, one of the patterns on the right is implemented.



Description	Characteristics
Simple order	A booking for a single Traveller with a transport provider, containing a trip from a pickup node to a drop-off node.
Shared transport	The transport route contains multiple – often overlapping – Travellers, each with a pickup and drop-off node. The route is built either Order by Order or as a DriverSession containing all Orders on the route.
Node by node transport	Most often a shared transport that is built dynamically during transport, one Node at a time.
Flag-stop	A Traveller signals that it wants to be picked up along the route. An Order is then created by the Provider.
Repetetive order	Repetetive Order: An Order that is carried out on a regular basis, booked according to a schedule (e.g. school transport).

Message format

SUTI messages shall conform to a format described in the standard documentation. Schema files (for XML and JSON) define the format and can be used to validate that messages are well-formed.

The top element of a SUTI Message is <SUTI>, immediately followed by `<orgSender … />` and `<orgReceiver … />`. The example shows `orgSender` and `orgReceiver` for a message sent by the Client. If the Provider sends the message, its `idOrg` will be in the `orgSender` section. The `orgSender` and `orgReceiver` id's are unique for each SUTI implementation.

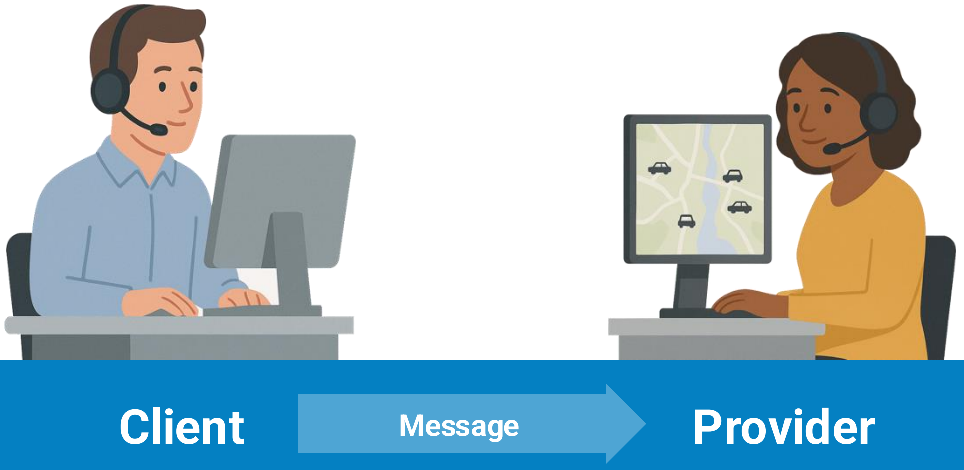
The example shows how the id-attribute of `idOrg` is built.

When initiating a new SUTI Link, the formats for all types of id are communicated between the Client and the Provider in a Link Mapping process, described in chapter 3 of How to use SUTI.

The `clientsw` and `providersw` names must be approved by the SUTI Technical Committee (TC), and the companies need to be members of SUTI for the SUTI Link to be considered an official implementation of the SUTI standard.

```
<SUTI>
  <orgSender name="clientname">
    <idOrg src="SUTI:idLink"
           id="clientsw_clientname_0024"
           unique="true"/>
  </orgSender>
  <orgReceiver name="providername">
    <idOrg src="SUTI:idLink"
           id="providersw_providername_0009"
           unique="true"/>
  </orgReceiver>
  …
  …

</SUTI>
```



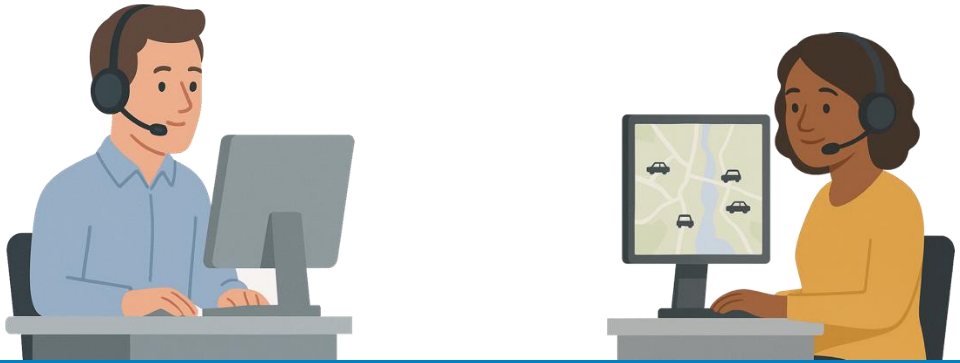
A simple Message

The example shows a simple but complete SUTI Message. A Message type in SUTI is identified by a four digit number and a corresponding name, found in the `msgType` and `msgName` attributes.

The example shows an Order confirmation by the Provider, containing a unique id for the message in `idMsg` and references to:

- the Order id in the Provider system
- the corresponding Order id previously sent by the Client system and
- The id of the Order Message sent by the Client, to which this confirmation is a reply.

```
<SUTI>
  <orgSender name="Provider name">
    <idOrg src="SUTI:idlink"
           id="providersw_providername_0009"
           unique="true"/>
  </orgSender>
  <orgReceiver name="Client name">
    <idOrg src="SUTI:idlink"
           id="clientsw_clientname_0024"
           unique="true"/>
  </orgReceiver>
  <msg msgType="2001" msgName="Order confirmation">
    <idMsg src="providersw_providername_0009:idMsg"
           id="20200915405027" unique="true"/>
    <referencesTo>
      <idOrder src="providersw_providername_0009:idOrder"
               id="2068608-0"/>
      <idOrder src="clientsw_clientname_0024:idOrder"
               id="482" unique="true"/>
      <idMsg src="clientsw_clientname_0024:idMsg"
             id="20200915822999" unique="true"/>
    </referencesTo>
  </msg>
</SUTI>
```



Client ← Order confirmation **Provider**

An Order Message

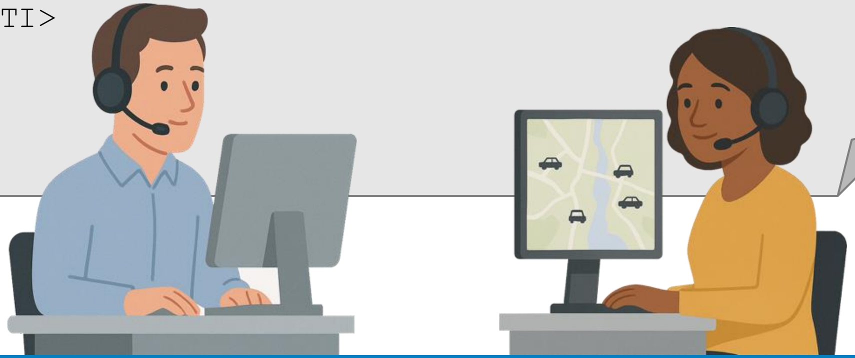
This XML example shows the general structure of an Order sent by a Client.

Within the `<Order>` element a number of subelements can be found:

- All Orders have a unique id – `idOrder`.
- The Provider is identified via `orgProvider`.
- The agreement that is the basis for the transport is defined by the agreement element.
- The `process` element contains a description of the transport process.
- `economyOrder` specifies the specific economic terms for the Order, e.g. the Price of the transport.
- `resourceOrder` specifies the requirements on the Driver and Vehicle to fulfil the Order.
- The `route` contains all the Nodes that needs to be visited to complete the Order.
- Each Node contains the geographical location of the Node, the Contents associated with the Node and the type of action to be performed at the node (e.g. pickup, destination).

This introduction will not cover the details of more complex Order types. This information can be found in the full documentation of the SUTI standard.

```
<SUTI>
  …
  <msg msgType="2000" msgName="Order">
    <idMsg … />
    <referencesTo>
      <order>
        <idOrder … />
        <orgProvider … />
        <process …/>
        <economyOrder … />
        <resourceOrder> … />
        <route>
          <node … />
          …
        </route>
      </order>
    <referencesTo>
  </msg>
</SUTI>
```



Client → Order → Provider

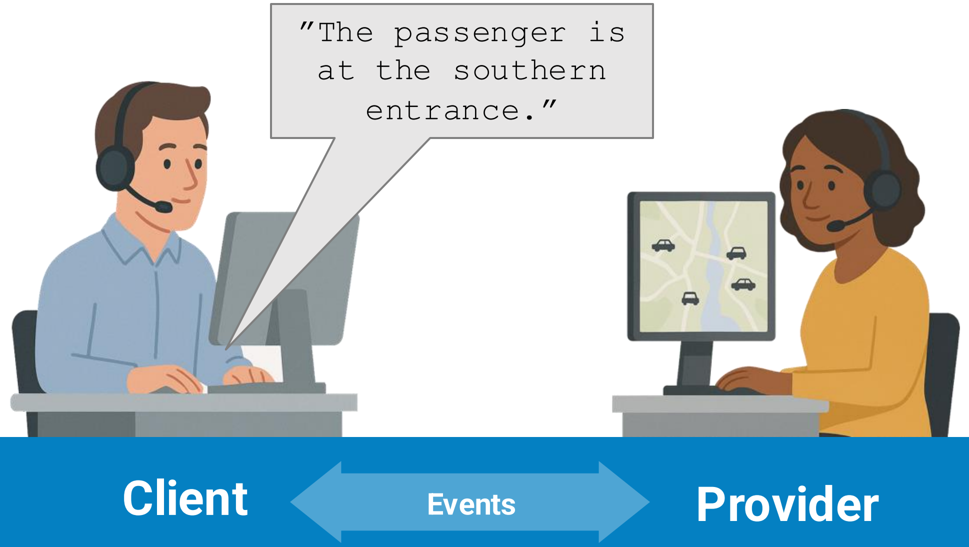
Events and text messages

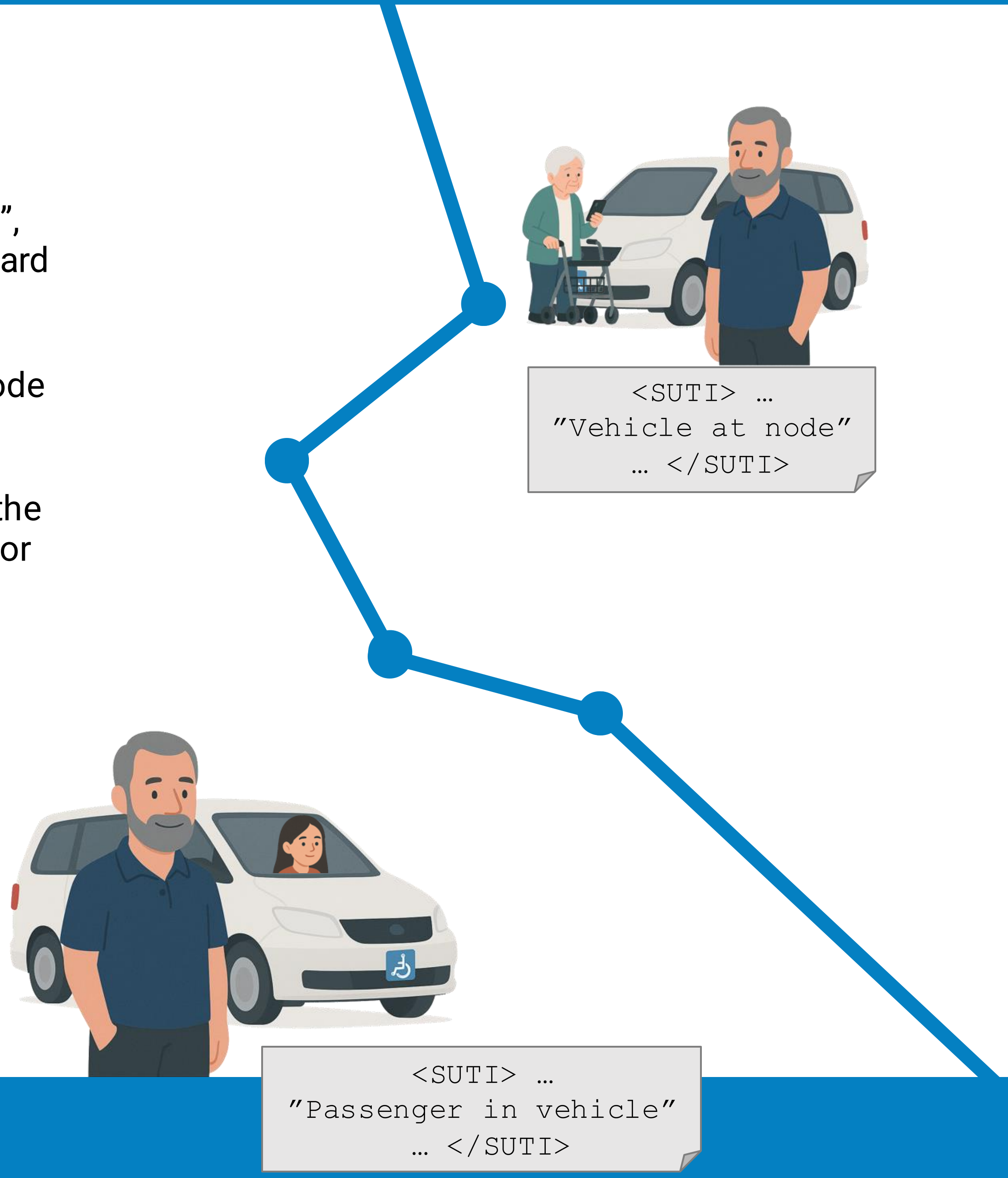
The Provider can report actions performed at a Node back to the Client as Events using a SUTI Message.

Different actions correspond to different eventTypes, e.g. ”Vehicle at Node”, ”Passenger in Vehicle”, etc. All eventTypes are described in the SUTI standard documentation.

It is also possible for the Client to inform the Provider about Events at a Node using a designated Message Type.

In addition it is possible to exchange customized text messages between the Client and the Vehicle (Driver) during transport, for communicating details or information that can not be conveyed via standard Message Types.





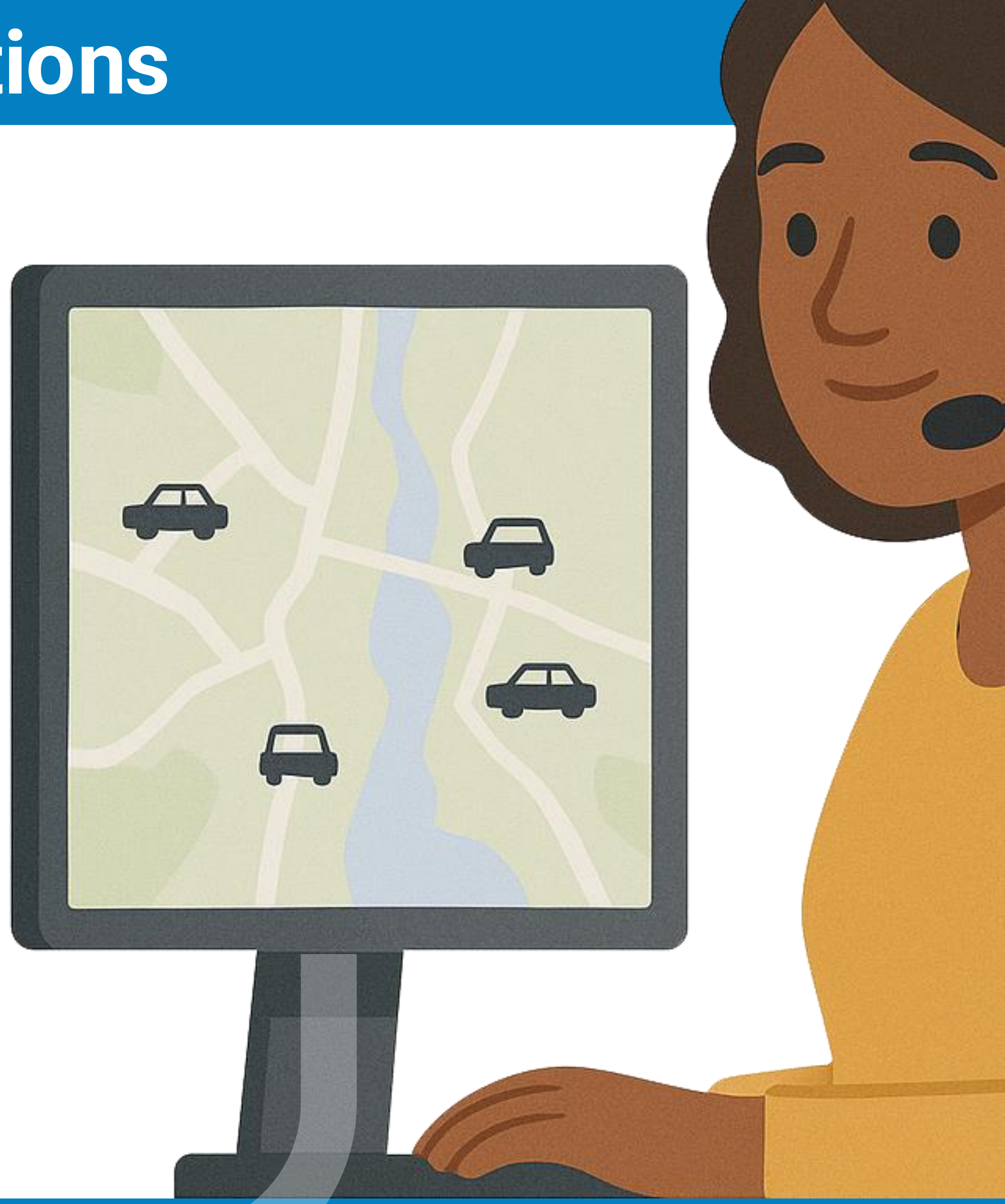
Vehicle locations

A common use case is that the Client needs information about the geographical location (GPS-position) of Vehicles. There are two main methods to achieve this.

The simple method is where Vehicle coordinates are requested for a specific Order. This can be either a one-time request or a request for coordinates to be sent at a set interval, for instance every 15 seconds.

If Vehicle coordinates are needed for many – or all – vehicles, requesting them per Order would result in verbose communication. Therefore there is an alternative method called Bulk Location Request, where all Vehicle locations are requested for a specific Agreement, compiled in one response message. This version also supports locations to be transmitted at a set time interval, e.g. 15 seconds.

Due to the higher bandwidth required by Bulk Location communication, it is commonly set up via a separate SUTI Link, in addition to the main Link where Order information is transmitted. Bulk Location communication can also be transmitted via JSON, thus minimizing communication load.



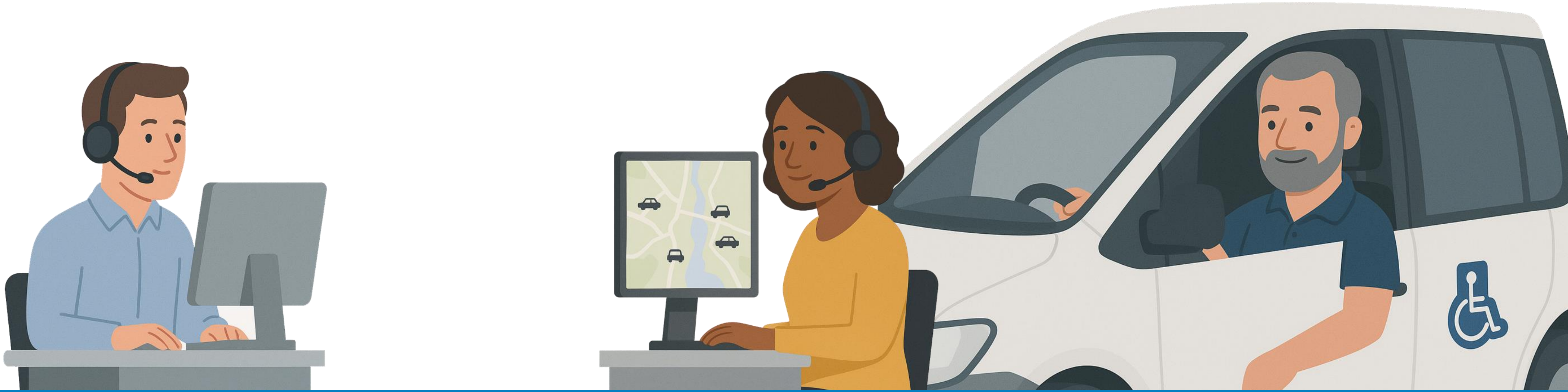
Client ← Bulk Location Response — **Provider**

Reporting

When an Order has been completed, the Client can request that the Provider sends an Order Report or a Delivery Report.

These reports contain economic information that can be used for calculating compensation as part of the accounting process. The Reports are gathered for all Orders within a time fram, e.g. a month or a week, and the total compensation is aggregated and calculated for that time frame.

The details of this communication is described in the Accounting section of the SUTI standard.



Client

Order Report

Provider

The Self Declaration

A SUTI implementation will cover specific parts of the standard and will define a specific order flow and transport control.

The Client specifies its implementation design choices in a **Self Declaration**, containing descriptions and examples of messages and the order flow.

Before it can be published as part of a Request for Proposal, a Self Declaration needs to be reviewed and approved by the SUTI Technical Committee.

Chapter 3 of *How to use SUTI* contains a more more in-depth description of how to create a Self Declaration.

Documentation and support

Full documentation of the SUTI Standard is provided via the SUTI webpage at suti.se and the SUTI GitHub Repository.

The documentation is divided into two parts: *How to use SUTI* and a *SUTI Messages* reference.

Support

SUTI Members can request assistance from the SUTI **Technical Committee** when creating a Self Declaration or implementing a SUTI Link.

Non-members are also welcome with general questions regarding the SUTI standard and when it’s suitable for application.

The Technical Committe holds weekly meetings and will respond to questions as quickly as possible, usually within two weeks.

The Technical Committee can be reached at tc@suti.se